



Ergonomic design of safety protective wearables in confined space operations

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ABSTRACT

This paper introduces a design methodology that integrates Systems Theoretic Accident Modeling and Process (STAMP) with ergonomics to address the limitations of current safety protective wearables in confined spaces, including incomplete functionality, low adaptability, and complex operation. The methodology begins with the development of a STAMP-Human Factors Control model to identify Unsafe Control Action (UCA) within confined spaces. Subsequently, the Analytic Hierarchy Process (AHP) is employed to prioritize these UCA. User demand hierarchy models are derived from questionnaire surveys and interviews, which are then analyzed using the Kano model to define requirements that enhance user satisfaction. Risk assessments of critical task postures under confined space operations are conducted using the JACK virtual simulation. The design of safety productive wearables is refined based on UCA, user requirements, and risk assessment outcomes. Finally, design optimization and human factor analysis are performed using JACK.

1. Introduction

A confined space is defined as an enclosed or semi-enclosed area with limited access and egress, encompassing only a specific region [1]. Within the chemical industry, the operating conditions in such restricted areas are particularly arduous; these spaces are typically cramped and poorly ventilated, with a high risk of rapid accumulation of hazardous gases. Operators are required to wear breathing respiratory masks, safety belts, and carry an array of protective, monitoring, maintenance, and other equipment when entering these spaces to perform tasks. The poor fit of wearable equipment to the human scale, the complexity and pain of wearing it, and the Work Related Musculoskeletal Disorders (WRMSD) [2] caused by poor operating postures will significantly impact the efficiency of emergency operations and personnel occupational safety.

Numerous scholars have conducted research to enhance efficiency and occupational safety within confined spaces. Yang [3] established wearable design principles based on research to ensure that the wearables satisfies both the physiological characteristics of rescue personnel and the actual requirements in distinct mine conditions. Xu et al. [4] developed a neck exoskeleton protection system for confined spaces that fits the requirements of ordinary operating postures. Wang et al. [5] investigated mine safety challenges and developed an intelligent safety

helmet for use underground. Zhang et al. [6] elaborated on the main application scenarios, structural characteristics, and precautions when using various types of safety protective wearables, aiming to provide selection references for practitioners in various industries, enhance the safety of safety protective wearables, and ensure the safety of workers' lives. Du et al. [7] designed an intelligent airbag-based high-fall-protection air-conditioned suit with multimodal sensing technology. This air-conditioned suit was capable of detecting human movement and precisely identifying potentially hazardous situations. Drury et al. [8] demonstrated that a lack of space during lifting duties reduces the effectiveness of all activities that require movement and influences the operator's physical strain. In their study of the spinal strains of manual workers executing activities, Scheff et al. [9] discovered that the effect of operating posture on the spine is crucial, even more important than the weight-bearing factor. Angelia et al. [10] showed that potential dangers to worker safety are pervasive on an industrial or construction site, thus they designed a wireless sensor integrated into construction site networks to monitor the health and safety of workers. Zhou et al. [11] have conducted much research on smart helmets to follow workers and their activities on construction sites. It is also recommended that construction workers wear a smart flexible helmet to enable continuous monitoring and safeguard them from health hazards while working. Ju et al. [12] wireless sensor network routing sensing approach was successfully

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designed, tested, and applied on smart helmets for air quality and hazardous event detection in the mining industry.

Current domestic and international research primarily concentrates on the design of safety protective wearables, with scant attention to the UCA of operators in confined space conditions and risks of operating postures. Consequently, this study analyzes the UCA in confined spaces by constructing a STAMP model, assesses the posture risk of confined space operations using JACK virtual simulation, and performs human factors analysis on safety protective wearables. Tailored to operator needs, a wearable equipment styling scheme is devised. Ultimately, the rationality of the scheme is verified through human-engineering interaction analysis and design evaluation in JACK simulation, addressing the issues present in the use of existing wearables during confined space operations.

2. Methods

Build a STAMP-Human factors control model to identify UCA in confined space, and use the AHP method [13] to weigh and analyze UCA. Combine the Kano model [14] to convert user demand hierarchy information, and evaluate critical task postures in confined spaces operations through JACK virtual simulation.

2.1. Analysis of UCA in confined space based on the STAMP model

In 2004, Leveson [15] introduced Systems Theoretic Accident Modeling and Process (STAMP), which encompasses various strategies to effectively identify UCA during the operation of complex systems. The process of system failure, attributed to the failure or inadequacy of safety control structures in confined space operations, is inherently complex. The STAMP model facilitates the identification and categorization of unsafe behaviors within confined spaces, and enables the screening of potential UCA that could arise during interactions with safety protective wearables throughout the workflow. Initially, safety constraints are applied to curtail operations posing risks; subsequently, a hierarchical control structure is instituted to regulate control behaviors more systematically; and ultimately, the process model is scrutinized to achieve a more profound comprehension of the operational mechanisms and potential risks of control behaviors[16]. Based on the STAMP model, Leveson introduced a new hazard analysis method, Systems Theoretic Process Analysis (STPA) [17], which centers on identifying UCA in the control feedback structure and sorting out the reasons for the occurrence

of UCA and restricting them.

2.1.1. System security constraint identification

To ensure the proper conduct of confined space operations, safety constraints include: Before operation, operators must verify that each other's protective wearables are fully donned to ensure safety; During operation, they must keep wearing the protective wearables and use the breathing respiratory correctly; After operation, maintenance personnel are required to perform scheduled wearables inspections and preventive maintenance.

2.1.2. STAMP-human factors control model

According to the process related to the operation of the confined space, the safety protective wearables were integrated into the STAMP basic model, and based on the STPA analysis method, the STAMP-Human factors control process model for safety protective wearables in confined spaces was developed as illustrated in Fig. 1. The illustration shows that safety protective wearables and the feedback mechanism of operation satisfaction are mapped to the actuator and transducer roles in the process model, respectively [18]. Within this framework, the operator group assumes the pivotal role of the controller, executing confined space operations while wearing safety protective wearables. Upon completion of the operation, the feedback regarding the use of safety protective wearables are conveyed to the operator team, thereby completing the feedback loop.

2.1.3. Identification of UCA

When conducting an in-depth review of the various components in the control loop, guided by the STPA analysis method, a detailed analysis is conducted on the potential UCA that may exist when operator interactions between operators and safety protective wearables throughout the entire workflow. These unsafe behaviors are classified into four types of control defects, with potential unsafe behaviors corresponding to various interaction factors within the human factors system, thereby offering a basis for subsequent optimization design. The identification results are shown in Table 1.

2.1.4. Screening for UCA

During the design process of safety protective wearable ewearables, a thorough quantitative analysis of potential UCA is essential. The purpose is to screen out the key control behaviors that account for a large proportion and carry out targeted design optimization for these behaviors.

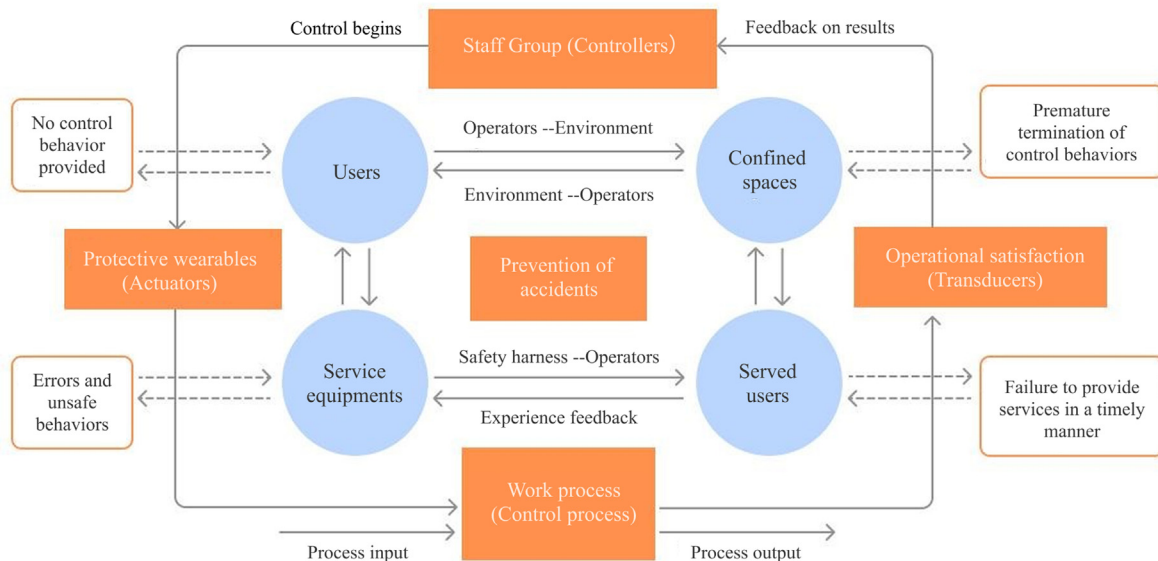


Fig. 1. STAMP- Human factors control model for safety protective wearables.

Table 1
Unsafe Control Action.

Types of UCA	UCA	Course of events
1. Behavior is not provided	1.1 Wearable equipment cannot provide services	No.1 Before the operation, the operator did not check the wearable equipment
	1.2 Unable to use wearable equipment for work	No.2 The wearable equipment malfunctions during use No.3 Unexpected conditions such as toxic gases make it impossible to provide services
2. Unsafe Control Action	2.1 Wearable equipment with inherent defects	No.4 The inconvenience of wearing equipment leads to low work efficiency No.5 Wearing equipment makes the worker's body load great
	2.2 Operator have UCA	No.6 Failure to use the wearable equipment in accordance with the operating guidelines No.7 The job was not carried out in accordance with the correct process
3. Control behavior delays	3.1 Delay in the use of wearable equipment during operation	No.8 The wearing time of the equipment is too long No.9 It takes a long time to check whether the equipment is safe to wear No.10 Try to resolve the glitch
		No.11 Inconvenience during the descent with wearable equipment
4. The act ends prematurely	4.1 Early termination of the use wearable equipment in the course of operations	No.12 Working with equipment for long periods leads to muscle fatigue No.13 The job is difficult No.14 The environment does not allow for long working hours

For the 14 possible behavior processes in Table 1, the AHP was employed to establish the hierarchical structure [19]. The significance of UCA in the objective layer is denoted as A, the criteria layer as Bi (i = 1, 2, 3), and the elemental layer as Ci, as depicted in Fig. 2.

To ensure the objectivity of UCA evaluation process, an expert panel comprising safety protective wearable designers, confined space operators, and design specialists was convened. The 1–9 Scale method was employed as the scoring criteria, as detailed in Table 2.

According to the scores of experts, the judgment matrix was constructed, and the analysis was as follows:

Table 2
1–9 Scale method.

Scale	Statement of importance
1	Equally important
3	Row indicators are slightly more important than column indicators
5	Row indicators are more important than column indicators
7	Row indicators are very important for column indicators
9	Row indicators are absolutely vital to column indicators
2,4,6,8	Two of the above judge the importance of the middle
Reciprocal	If A is more important than B, then row A and column B are labeled 5; when analyzing the importance of B to A, fill in 1/5

$$A = \begin{bmatrix} 1 & 3 & 3 & 8 \\ 1/3 & 1 & 2 & 5 \\ 1/8 & 1/5 & 1/4 & 1 \\ 1/3 & 1/2 & 1 & 4 \end{bmatrix} \quad B_1 = \begin{bmatrix} 1 & 3 & 5 \\ 1/3 & 1 & 2 \\ 1/5 & 1/2 & 1 \end{bmatrix} \quad B_2 = \begin{bmatrix} 1 & 3 & 3 & 4 \\ 3 & 1 & 2 & 5 \\ 8 & 6 & 1 & 4 \\ 3 & 1/2 & 7 & 4 \end{bmatrix}$$

$$B_3 = \begin{bmatrix} 1 & 1/2 & 3 & 4 \\ 2 & 5 & 2 & 1/5 \\ 1/3 & 6 & 1 & 4 \\ 6 & 1/2 & 7 & 1 \end{bmatrix} \quad B_4 = \begin{bmatrix} 1 & 5 & 7 \\ 1/3 & 1 & 3 \\ 1/7 & 1/3 & 1 \end{bmatrix}$$

The YAAHP software was utilized to calculate the judgment matrix and verify the results that met the consistency test criteria. Subsequently, YAAHP was employed to calculate the weight value of each element in

Table 3
Combined weights of potential UCA for safety wearable equipment.

Guideline layer weights	The weighting of factors at the decision-making level		Combined weights	Ranking
(B ₁)0.295	(C ₁)	0.231	0.215	2
	(C ₂)	0.132	0.071	6
	(C ₃)	0.083	0.028	11
(B ₂)0.539	(C ₄)	0.444	0.239	1
	(C ₅)	0.750	0.083	4
	(C ₆)	0.666	0.041	8
(B ₃)0.055	(C ₇)	0.240	0.071	5
	(C ₈)	0.129	0.032	10
	(C ₉)	0.136	0.040	9
(B ₄)0.168	(C ₁₀)	0.069	0.011	13
	(C ₁₁)	0.095	0.051	7
	(C ₁₂)	0.489	0.144	3
	(C ₁₃)	0.229	0.011	12
	(C ₁₄)	0.122	0.006	14

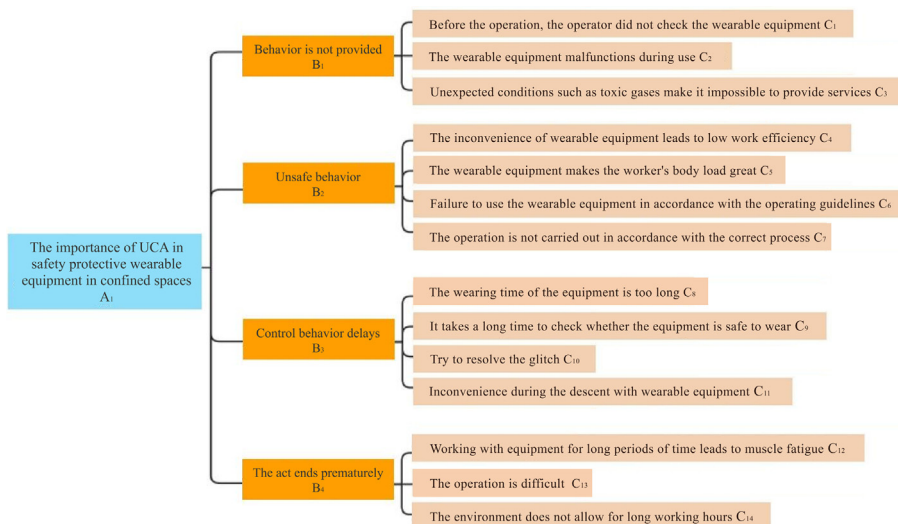


Fig. 2. Hierarchical model of UCA of safety protective wearables.

the scheme layer to derive the comprehensive weight of potential UCA of safety protective wearables, as shown in Table 3. These rankings guided the subsequent research.

2.2. Requirements analysis based on the Kano model

Noriaki Kanoproposed the Kano model is widely used in user demand acquisition analysis and adeptly captures the relationship between product performance and user satisfaction. Through the construction of the Kano model, the performance required for the safety protective wearables in confined spaces can be more effectively delineated, thereby significantly enhancing user satisfaction.

Initially, we targeted 36 confined space operators and managers to gather their original requirements via semi-structured questionnaires and interviews. We visualized user behavior paths using the user journey diagram, and comprehensively summarized user needs based on their psychological feelings. The results are shown in Table 4.

The KJ method [20] is used to organize and summarize the requirements, merge similar requirements, and form four categories of primary requirement groups, namely functional requirements, human-machine requirements, psychological requirements, and styling requirements. Based on the hierarchical order, we constructed the demand hierarchical model based on safety protective wearables in confined space, as shown in Fig. 3.

Utilizing the hierarchical model of user requirements and the Kano model analysis method, 5 confined space managers and operators were selected respectively to conduct a questionnaire survey on the satisfaction degree of safety protective wearable design in confined spaces. After completing the questionnaire survey, the collected data were systematically organized and determined the types of quality attributes based on the evaluation criteria of the Kano model. These types include essential quality elements (M), expected quality elements (O), excitatory quality elements (A), indifference quality elements (I), reverse quality elements (R), and problem quality elements (Q). The results of data analysis and organization are shown in Table 5.

By the frequency formula:

$$\text{Better} = (A + O) / (A + O + M + I) \quad (2-1)$$

$$\text{Worse} = -(O + M) / (A + O + M + I) \quad (2-2)$$

Eqn 2-1 and 2-2

The Worse and Better coefficients were obtained, subsequently, the Worse coefficient was used as the horizontal axis and the Better

coefficient was used as the vertical axis to analyze the results by using the visualized scatter plot, as shown in Fig. 4. Based on the scatter plot results, in the design of the safety protective wearable in the confined space, the contents of F_1 Multiple hanging points, E_1 Easy access to tools, E_2 Breathing apparatus fit, S_3 Lightweighting of equipment, P_1 Sense of security, P_2 Comfortableness and so on should be emphasized in the design. And as far as possible to meet the requirements of E_3 Leg cushioning, E_4 Back cushioning, E_6 Adaptation to the environment, S_2 Environmental compatibility, and S_4 Bright colors. Meeting the requirements of product S_5 Humanization, F_2 Precise positioning, F_3 Voice intelligence, F_4 Real-time monitoring and so on can greatly improve user satisfaction.

2.3. Risk assessment of confined space operating postures

2.3.1. Construction of digital mannequin and environment

The primary focus of this study was on Chinese adult male operators of safety protective wearables in confined spaces. In accordance with the national standards [21], the Chinese human body database was referenced in JACK to generate the 50th percentile Chinese male digital human model. It is shown in Fig. 5.

To enhance data accuracy, the survey workers carried objects with a load, including safety helmets weighing 430 g, breathing respiratory masks weighing approximately 500g, safety protective wearable weighing 2.8kg, two large double hooks weighing 2 kg, gas detectors weighing 480g, and walkie talkies weighing 320 g. The model size and positional relationships were adjusted according to simulation needs; forces were applied to the back, shoulders, waist and legs, while a 2 kg force (simulating maintenance tools) was applied to the hands, as shown in Fig. 6.

2.3.2. Risk assessment of critical task postures

Field observations and researchrevealed that in the specific field of confined space operations, narrow space has become a significant feature. In these environments, workers usually need to maintain four fixed postures: standing overhaul posture, tool changing posture, semi-squatting overhaul posture, and squatting overhaul posture. The Task Script Block (TSB) module in JACK was used to analyze the virtual simulation scene of maintenance work in the tank. The Rapid Upper Limb Analysis(RULA) and Ovako Working Posture Analysis System (OWAS) in TSB were used to assess the risk of critical task posture [22].

RULA can effectively evaluate the risk posed by working posture to the upper limb musculoskeletal system according to multiple dimensions such as human posture, muscle stress status, working task, and working time; it also provides a scientific basis for optimizing working environment, reducing working fatigue and preventing musculoskeletal diseases [23]. The meaning of each level in the RULA results are shown in Table 6.

OWAS work posture analysis plays a pivotal role in evaluating the comfort of working postures, as it facilitates the swift identification of the potential injury risks caused by these postures. Consequently, it helps determine the urgency of taking corrective measures [24]. The evaluation results are divided into four levels, and the meanings of the specific levels are shown in Table 7.

When conducting RULA, due to the existence of two states of the upper limb, namely, working and static, in the actual operation process [25], two sets of tasks were created:Group A of working postures and Group B of static postures, and corresponding muscle usage and load conditions were set. In group A, the use of human muscles was set to repeat more than 4 times per minute, and the load was intermittent and greater than 2 kg. In group B, the muscle use was set to maintain a static posture for more than 1 minute, and the load was also greater than 2kg. The scores of Group A and Group B were then integrated to obtain the overall score. According to RULA and OWAS, the risk assessment of four critical task postures including standing overhaul posture, tool changing posture, semi-squatting overhaul posture, and squatting overhaul posture are carried out in turn and a virtual simulation of JACK is constructed, as shown in Figs. 7–10, and the specific analysisresults are shown in Table 8.

Table 4
User requirements.

Number	Requirement
1	Multiple hanging points
2	Precise positioning
3	Voice intelligence
4	Real-time monitoring
5	Booster device
6	Human-computer interaction
7	Easy access to tools
8	Breathing apparatus fit
9	Leg cushioning
10	Back cushioning
11	Comfortable position
12	Adaptation to the environment
13	Appearance harmonization
14	Environmental compatibility
15	Lightweighting of equipment
16	Bright colors
17	Humanization
18	Sense of security
19	Comfortableness
20	Relieve stress
21	Reduce fear

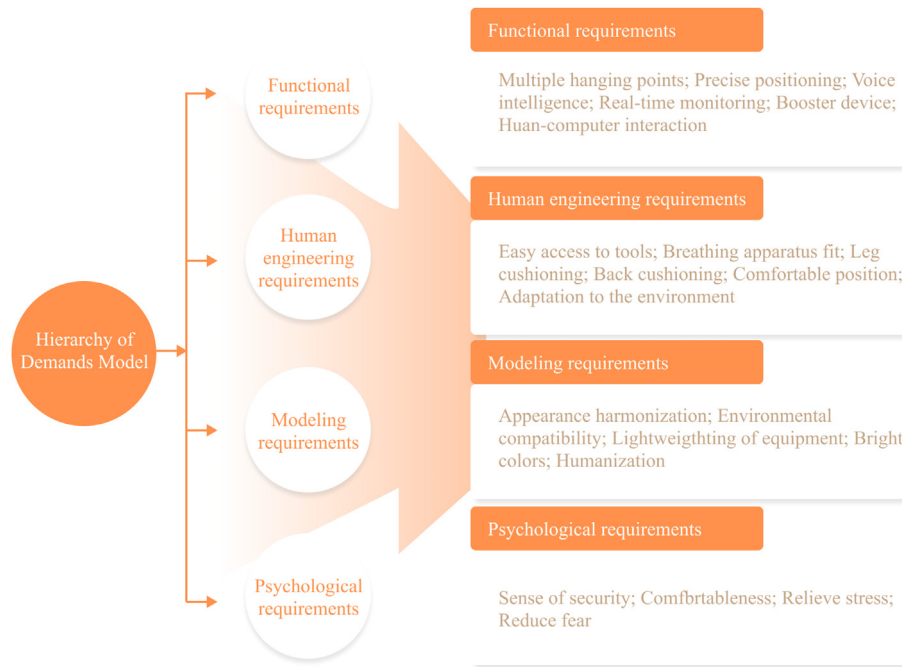


Fig. 3. The demand hierarchical model of safety protective wearables in confined spaces.

Table 5
Results of the questionnaire.

Requirement	Number	Demand indicators	M(%)	O (%)	A (%)	I (%)	Kano causality	Better(%)	Worse(%)
Functional requirements (F)	F ₁	Multiple hanging points	35.48	19.35	22.58	22.58	M	0.42	−0.55
	F ₂	Precise positioning	25.81	12.90	29.03	25.81	A	0.45	−0.41
	F ₃	Voice intelligence	22.58	16.13	35.48	19.35	A	0.68	−0.55
	F ₄	Real-time monitoring	25.81	22.58	32.26	16.13	A	0.65	−0.68
	F ₅	Booster device	19.35	25.81	22.58	32.56	I	0.48	−0.45
	F ₆	Human-computer interaction	25.81	9.68	25.81	35.48	I	0.37	−0.37
Engineering requirements (E)	E ₁	Easy access to tools Breathing apparatus fit	29.03	25.81	22.58	19.35	M	0.50	−0.57
	E ₂	Leg cushioning	32.26	22.58	25.81	16.13	M	0.50	−0.57
	E ₃	Back cushioning	25.81	41.94	19.35	12.90	O	0.61	−0.68
	E ₄	Comfortable position	22.58	41.94	19.35	16.13	O	0.61	−0.65
	E ₅	Adaptation to the environment	35.48	35.48	6.45	22.58	M	0.40	−0.70
	E ₆	Humanization	25.81	41.94	22.58	9.68	O	0.65	−0.68
Sculpt Requirements (S)	S ₁	Appearance harmonization Environmental compatibility	29.03	25.81	19.35	22.58	M	0.47	−0.57
	S ₂	Lightweighting of equipment	22.58	41.94	19.35	16.13	O	0.61	−0.65
	S ₃	Bright colors	32.26	29.03	12.90	25.81	M	0.42	−0.61
	S ₄	Humanization	16.13	35.48	22.58	25.81	O	0.58	−0.52
	S ₅	Humanization	22.58	32.26	35.48	9.68	A	0.57	−0.55
Psychological Requirements (P)	P ₁	Sense of security	35.48	16.13	16.13	32.26	M	0.48	−0.52
	P ₂	Comfortableness	32.26	25.81	22.58	19.35	M	0.48	−0.61
	P ₃	Relieve stress	22.58	9.68	25.81	41.94	I	0.35	−0.32
	P ₄	Reduce Fear	35.48	16.13	22.58	25.81	M	0.39	−0.52

As shown in Table 8, based on the RULA, the total scores for standing overhaul posture, tool changing posture, and semi-squatting overhaul posture are all 6 points, corresponding to a posture evaluation level of 3. This indicates that these postures have adverse effects on the physical health of workers and require prompt corrective or improvement measures. The total score of the squatting overhaul posture is 7 points, corresponding to the assessment level of 4, indicating that the squatting overhaul posture significantly adversely affect the physical health of two workers and requires immediate corrective measures. According to OWAS, the correction requirement level for standing overhaul posture and tool exchanging posture is 1, indicating that these postures are normal and do not require special correction or improvement. The correction requirement level for semi-squatting overhaul posture is 2, indicating that this posture may have harmful effects on the musculoskeletal system, but its muscle load is not extreme, and corrective

measures are encouraged. In contrast, the correction requirement level of the squatting overhaul posture is 3, indicating that the squatting overhaul posture has harmful effects on the musculoskeletal system and requires to be corrected or improved immediately.

2.4. Design of safety protective wearables

2.4.1. Sketch

Combined with the weight ranking of UCA, user requirements, and the risk assessment results of critical task posture, this study aimed to design a complete set of safety protective wearable based on the existing safety harnesses and self-contained breathing apparatuses. The sketch scheme modeling were deduced, as shown in Figs. 11 and 12.

A quick snap fastening has been created for easy on and off. Cushions have been added to the shoulders, waist, and legs to relieve muscle

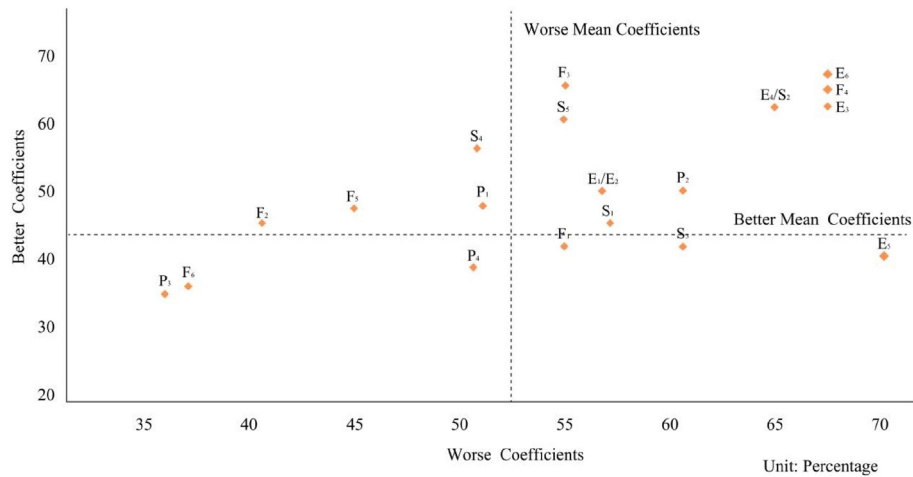


Fig. 4. Scatter plot.

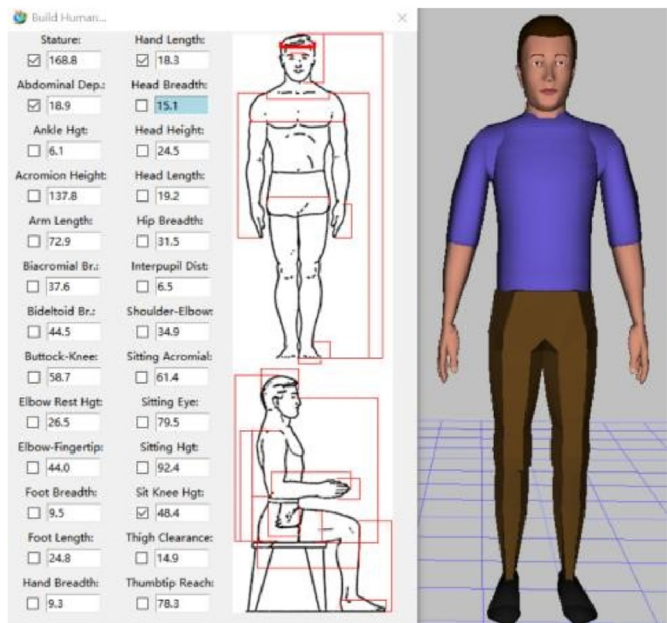


Fig. 5. The model represents the average physical characteristics of people working in confined spaces in China, with the height set at 168.8 cm and the weight at 59 kg.

tension on workers during the process. The self-contained breathing apparatus was a long tube respirator of air supply type, with double joints, which can be connected to an external air supply and escape gas cylinders. The overall sketch scheme is shown in Fig. 13.

2.4.2. 3D model scheme

This design includes two main parts: a safety harness and a breathing apparatus (equipped with gas cylinders, pressure gauges and high-pressure pipes), providing stable respiratory support for operators. The scheme is shown in Fig. 14. This wearable device can be combined with a double hook up and down, as shown in Fig. 15.

3. Results

3.1. Scheme simulation evaluation based on JACK

The 3D model was integrated into JACK and the design rationality was verified using the previously established environment, character

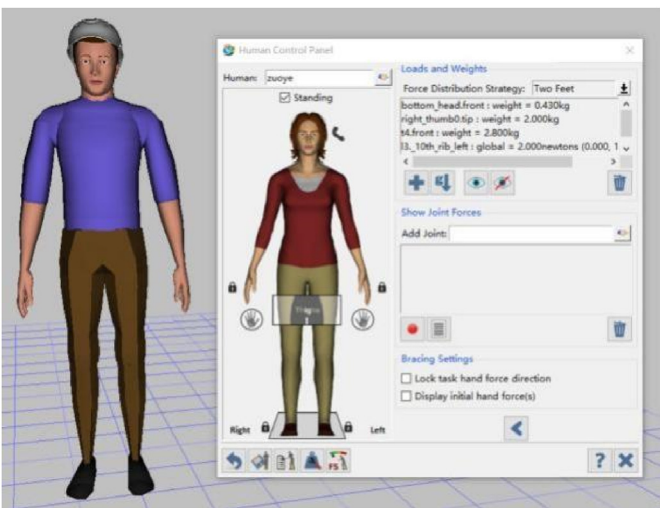


Fig. 6. Model size and position relationship.

Table 6

Results of RULA.

Rate	Level	Analysis
1-2 points	1	The posture is feasible and there are no significant risk.
3-4 points	2	This posture carries certain risks and requires action to study and improve.
4-6 points	3	The posture carries a high risk and requires action to study and improve as soon as possible.
7 points	4	The posture poses serious risks and requires immediate action to study and improve.

Table 7

Results of OWAS.

Level	Analysis
1	This posture falls within acceptable biomechanical ranges, and no special correction or improvement is required.
2	This posture may have some adverse effects on the individual, and correction or improvement is recommended in the short term.
3	This posture has adversely affected the individual, and corrections or improvements are recommended as soon as possible.
4	This posture has caused significant harm to the individual, and corrections or improvements must be implemented immediately.

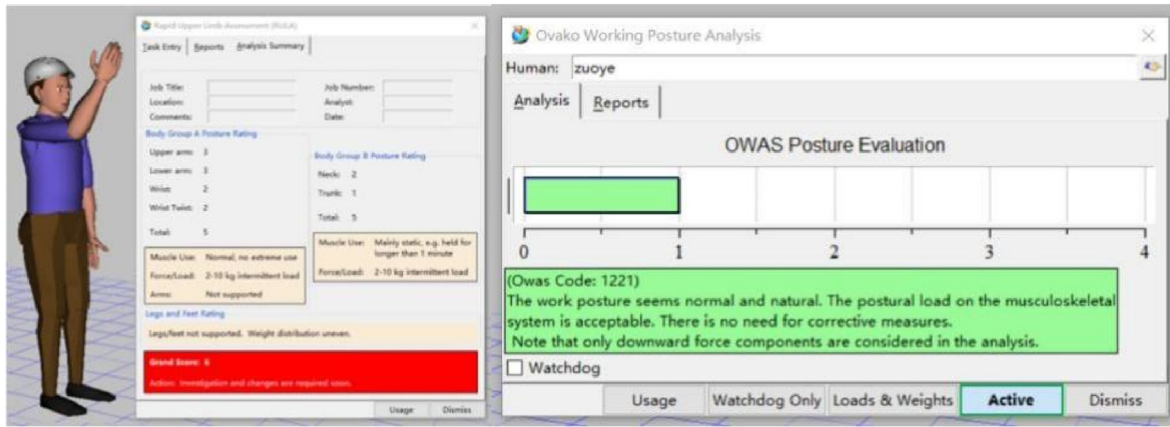


Fig. 7. RULA/OWAS based on JACK Virtual Simulation of standing overhaul posture.

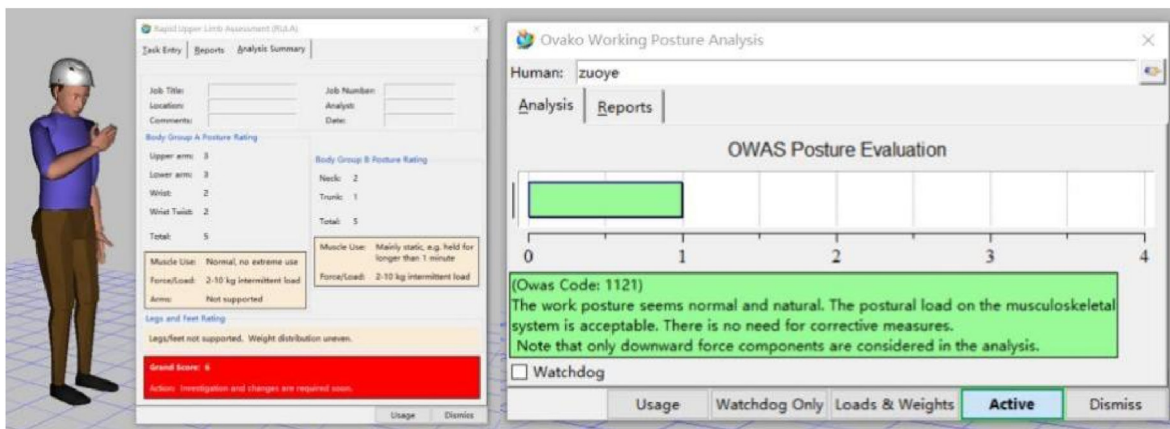


Fig. 8. RULA/OWAS based on JACK Virtual Simulation of tool changing posture.

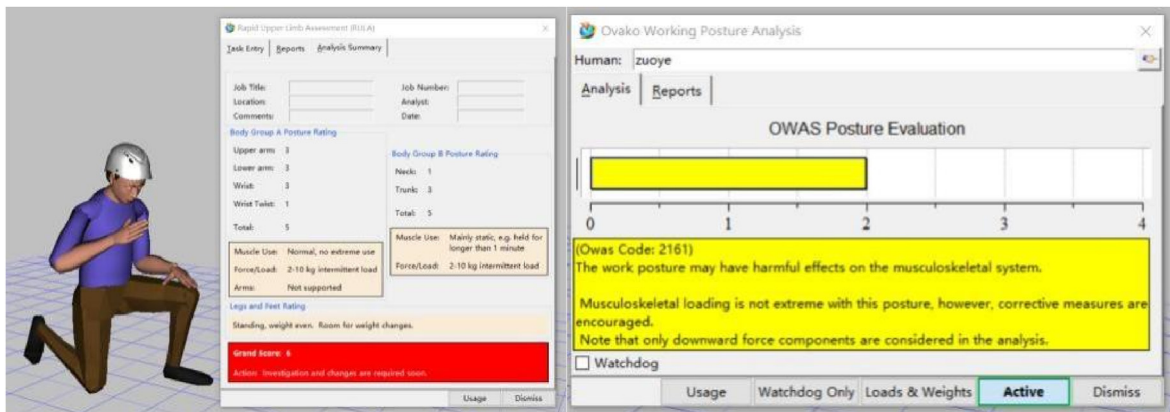


Fig. 9. RULA/OWAS based on JACK Virtual Simulation of semi-squatting overhaul posture.

model, and posture. Within JACK, the degrees of freedom of the joints of each part of the created digital human model were adjusted so that the model of the safety protective wearable remains in place during postures changes.

3.1.1. RULA of critical task postures

In conjunction with the product design scheme, the four critical task postures were subjected to RULA, with specific results depicted in Fig. 16. The RULA scores for the standing overhaul posture, tool changing posture, and semi-squatting overhaul posture decreased from 6 points to

3, while the squatting overhaul posture score decreased from 7 points to 3. These reductions indicate a significant decrease in stress load on the upper limbs and a more reasonable distribution of gravity.

3.1.2. OWAS of critical task postures

This section primarily examines the load conditions on the operator's back, arms and legs, and comprehensively assesses the safety risks of work postures. The detailed analysis results are shown in Fig. 17. The results show that the correction requirement for the standing overhaul posture and tool changing posture remained at level 1. In contrast, the

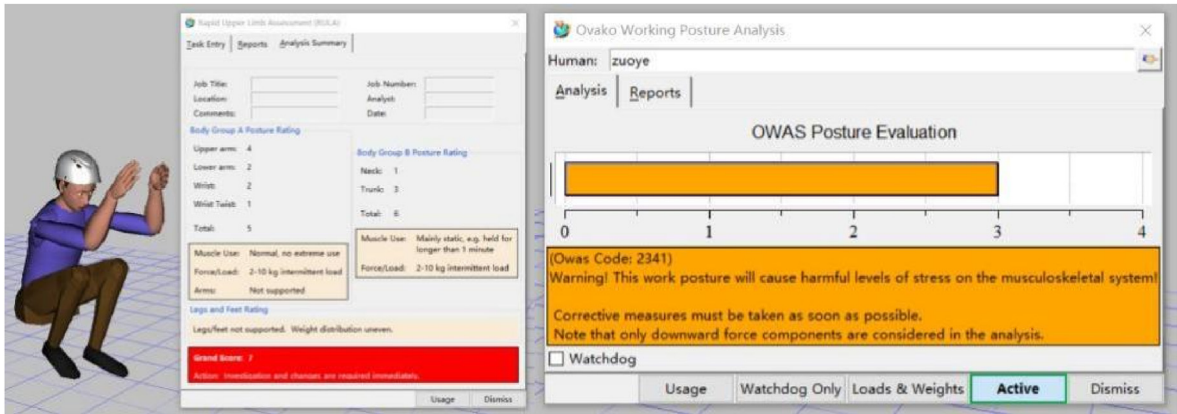


Fig. 10. RULA/OWAS based on JACK Virtual Simulation of squatting overhaul posture.

Table 8
Results of RULA/OWAS for four critical task postures.

Posturing	Methodology	Clusters	Position	Rate	Overall rating	Level
Standing overhaul posture	RULA	A	upper arm	3	6	3
			lower arm	3		
			finesse	2		
			wrist	2		
			flexion	2		
		B	neck	2		
Tool changing posture	RULA	A	upper arm	3	6	3
			lower arm	3		
			finesse	2		
			wrist	2		
			flexion	2		
		B	neck	2		
Semi-squatting overhaul posture	RULA	A	upper arm	3	6	3
			lower arm	3		
			finesse	3		
			wrist	1		
			flexion	1		
		B	neck	1		
Squatting overhaul posture	RULA	A	upper arm	4	7	4
			lower arm	2		
			finesse	2		
			wrist	1		
			flexion	1		
		B	neck	1		
	OWAS	/	trunk	3	/	2
				/		
	OWAS	/		/	/	3
				/		

correction requirement level for the semi-squatting overhaul posture has changed from level 2 to level 1, and for the squatting overhaul posture, it has been reduced from level 3 to level 1. This suggests that these postures have no adverse effects on the musculoskeletal system of the workers due to the optimized design of the safety protective wearable.

3.1.3. Comfort assessment

The comfort assessment tool, selected from the Occupant Packaging

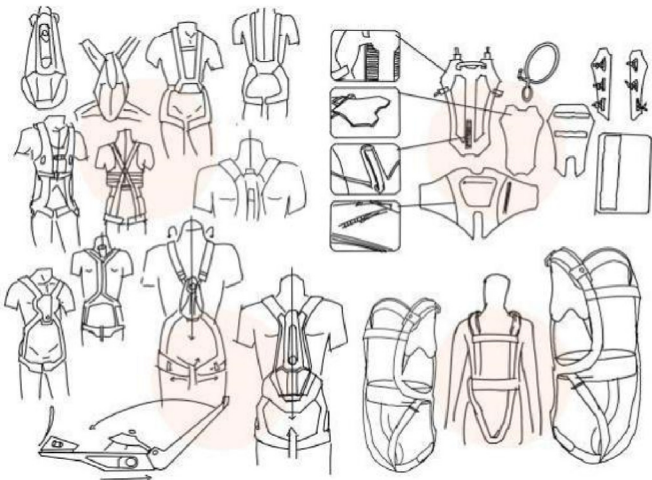


Fig. 11. Safety harnesses sculpt refinement.

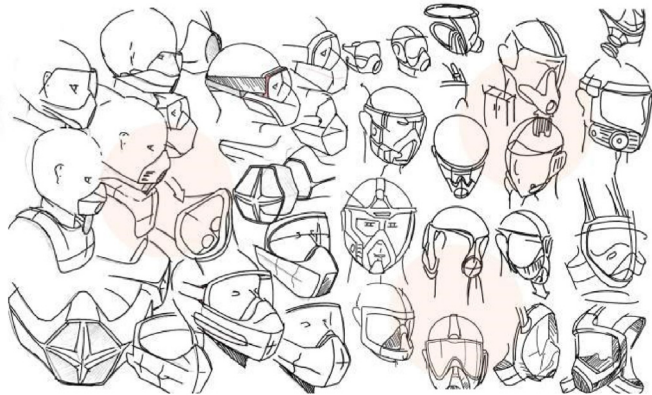


Fig. 12. Self-contained breathing apparatus sculpt refinement.

Toolkit (OPT) module within JACK, was utilized to assess the operator's posture comfort. This tool categorizes comfort data sources into single-joint comfort analysis and multi-joint comfort analysis. A multi-joint comfort analysis (Krist's model) was selected to comprehensively evaluate the whole-body comfort of four operating postures. Fig. 18 displays the final comfort analysis results.

The comfort analysis findings are quantitatively evaluated, with a value range of 0–80. These findings are presented as a unilateral bar chart, where the score is inversely proportional to comfort level; that is, a

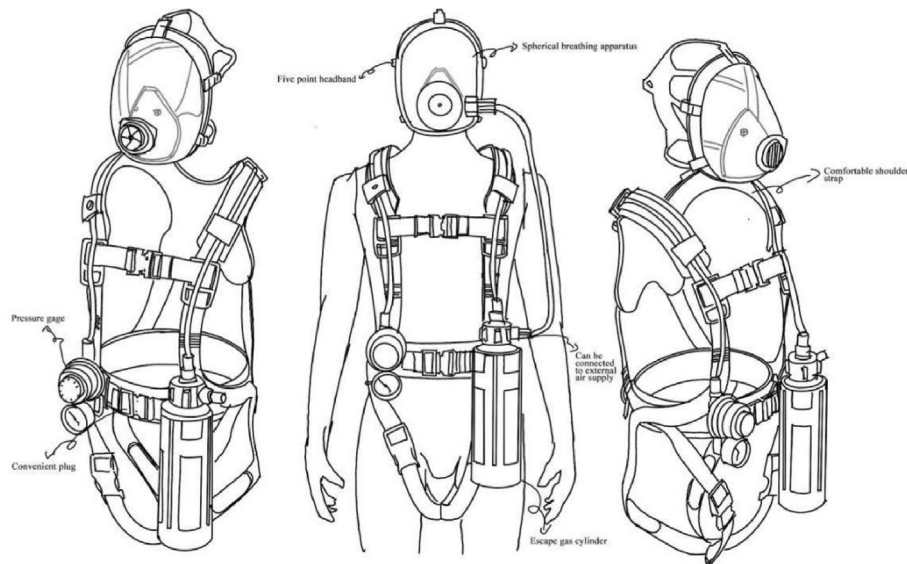


Fig. 13. Sketch scheme.

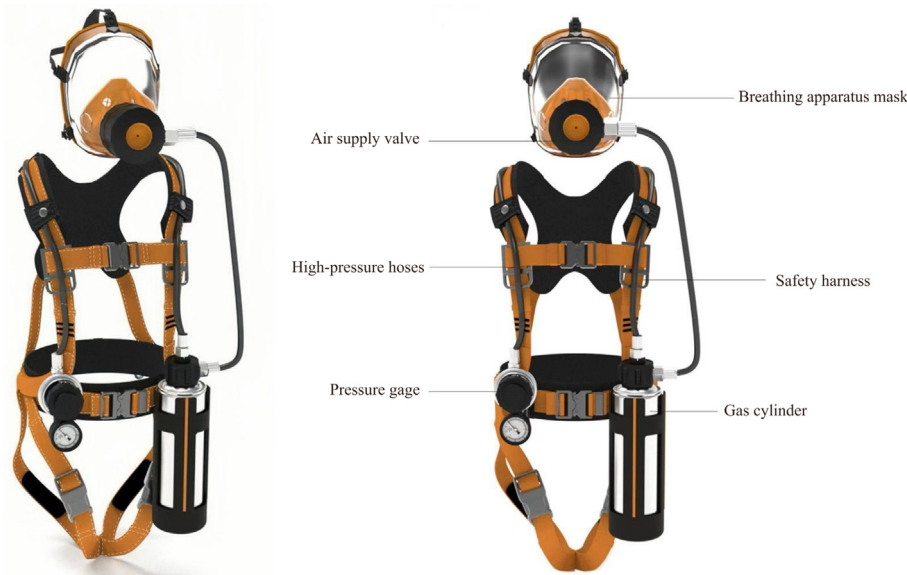


Fig. 14. 3D model scheme.

lower value indicates a higher comfort level. A yellow or red zone on the bar graph signifies a less comfortable posture; green signifies a posture within the comfortable range. The bar charts for the four posture assessments shown in the diagram are all in green, indicating that all body parts are in a comfortable state during work after the operator wears the safety protective wearable.

3.2. Human factors analysis and design optimization of the scheme

3.2.1. Lightweight human factors analysis and design optimization

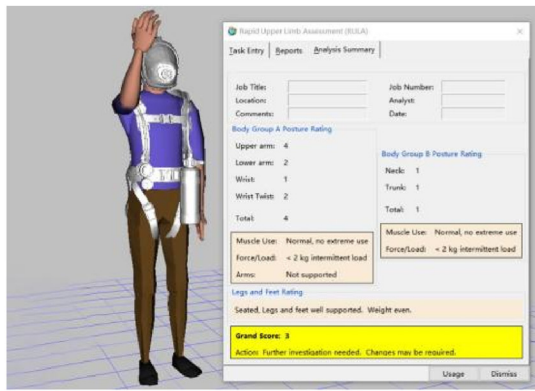
- (1) **Lightweighting material:** While ensuring safety performance, the weight of safety protective wearables should be minimized. Polyester and lightweight alloys were selected as the primary materials because they are lightweight, flexible, cost-effective, and long-lasting enough to be employed in the creation of safety protective wearable.

- (2) **Lightweighting structure:** Topology optimization, shape optimization, size optimization, and other procedures are among the lightweight strategies used in structural design [26]. Shape optimization refers to altering the design of components to improve their mechanical properties while fulfilling strength criteria to achieve lightweight [27].

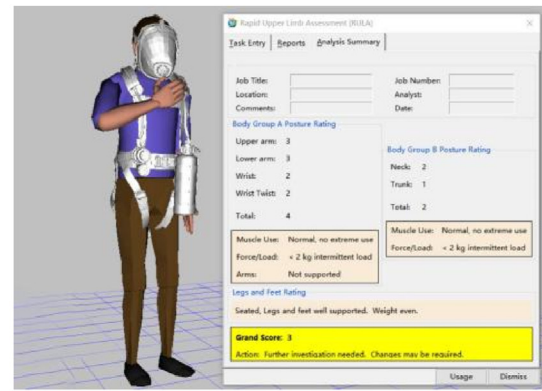
In accordance with the weight-bearing mechanism of safety protective wearables, the lumbar fossa above the sacrum and the connection between the lumbar vertebrae are identified as the main weight-bearing area, with the shoulders assisting in support, thus ensuring the balance and stability of the carrying system [28]. From the perspective of lightweight and breathability, it is necessary to minimize the number of support points as much as possible. Based on the principles of geometry, a three-point support structure is most stable under stress. Consequently, using a backpack structure with two shoulder supports and one waist support minimizes the strain on the spine while also achieving structural lightweight.



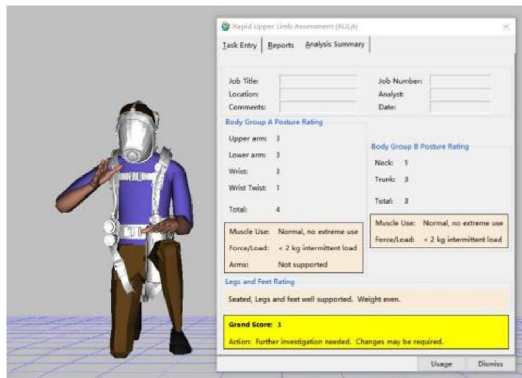
Fig. 15. Human-machine wearable renderings.



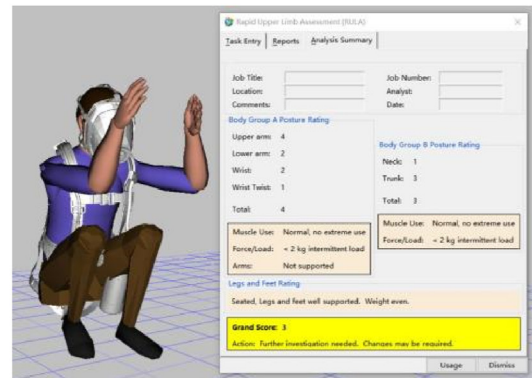
(a) standing overhaul posture



(b) tool changing posture



(c) semi-squatting overhaul posture



(d) squatting overhaul posture

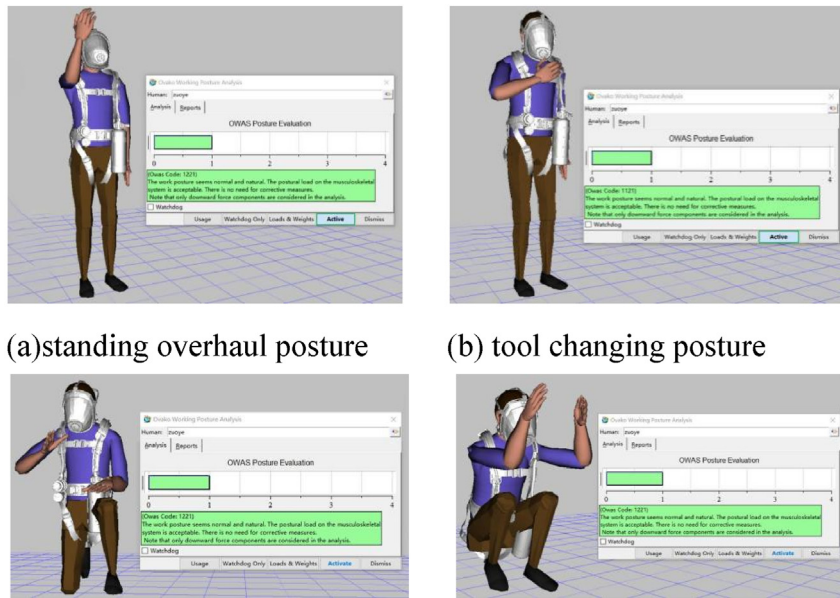
Fig. 16. RULA analysis of critical task postures.

3.2.2. Human factor analysis and design optimization of comfort

In addition to lightweight design considerations, the wearer's comfort should be carefully considered. The key elements involved in wearing safety protective wearables mainly include the rationality of the hanging point position and the implementation of an adjustable design.

- (1) Rationalization of hanging point position: The positioning of the hanging points will affect the operator's comfort, so it has been

improved. Due to its special nature, safety protective wearables requires the use of anti-fall devices, winches, and double hooks to ascend and descend, so there are special requirements for the hanging points. Concurrently, for stability, the overall wearable equipment features a five-point hanging design, with the waist serving as the primary load-bearing center [29]. Two hanging points on the chest work in tandem with one hanging point on the back to connect the fall arrest system, ensuring the operator's

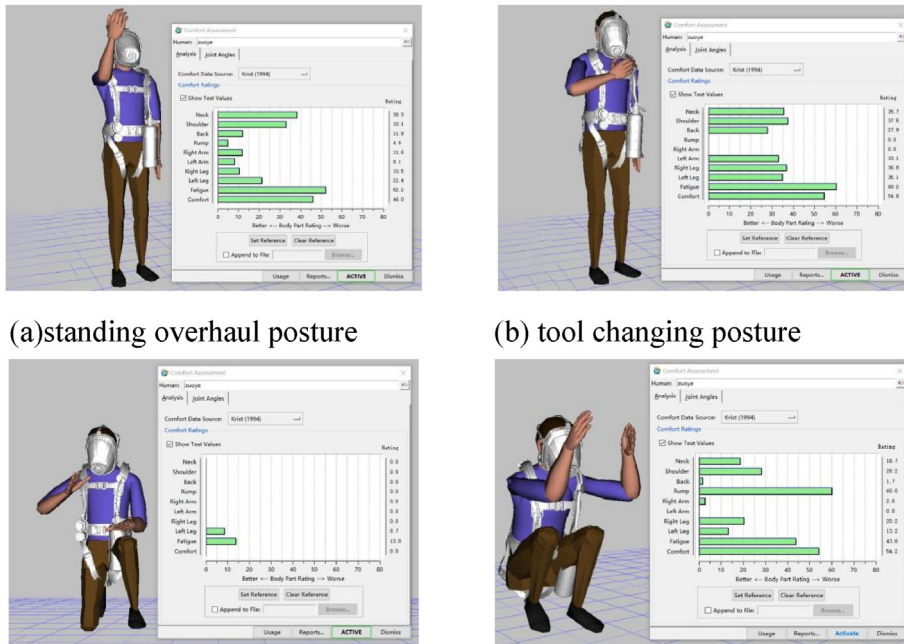


(a)standing overhaul posture

(b) tool changing posture

(c) semi-squatting overhaul posture (d) squatting overhaul posture Figure

Fig. 17. OWAS analysis of critical task postures.



(a)standing overhaul posture

(b) tool changing posture

(c)semi-squatting overhaul posture

(d) squatting overhaul posture

Fig. 18. Comfort assessment.

safety, while the hanging points on both sides of the waist connect the work limitation system, limiting the working range and preventing accidents. As indicated in Fig. 19.

- (2) Adaptable design: To ascertain whether the back belt and mask are fully worn before the operator commences work, thereby preventing accidents, the head belt, chest belt, waist belt, and leg belt adopted a quick-insert buckle system. Meanwhile, the adaptable design was used to accommodate the operators' varying heights and weights. By refining the adjustment mechanism, the safety protective wearable fits the operator's body and achieves

organic integration. Fig. 20 displays the adjustable construction, which primarily involves cushioning pad and strap adjustments.

The breathing respiratory mask adopts a five-point adjustable head-band, as shown in Fig. 21, which conforms to the shape of the head, has uniform weight distribution, and is more comfortable and safe. Given that the proportion of female employees in the confined space industry is less than 10 %, this study mainly refers to male head size, based on the national standard -GB 10000-1988 Chinese Adult Human Body Size. The data table of male head size for Chinese adults provides a reference for

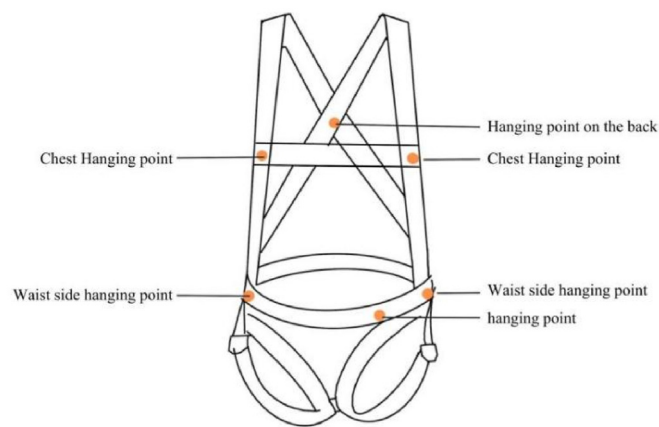


Fig. 19. Hanging point design.

the design of breathing respiratory masks. By referring to the size table, first calculate the mean and standard deviation. Considering the adjustability of the breathing respiratory mask, we select the 50th percentile data in the table, which is the head size of the medium-sized population, as the principal reference.

4. Discussion

In this study, the STAMP-Human factors control model was applied to develop the safety protective wearable in confined spaces, allowing for comprehensive consideration of accident prevention and comfort at the system level. By combining with the virtual simulation technology of JACK, it realized accurate analysis and shape optimization of the design of safety protective wearable , which improved the scientific and practicality of the product. The design concept of safety protective wearables in confined spaces was then provided, enriching design theory in related domains and providing guidance for safety protective wearables research



Fig. 20. Schematic diagram of the adjustable structure of the safety harness.

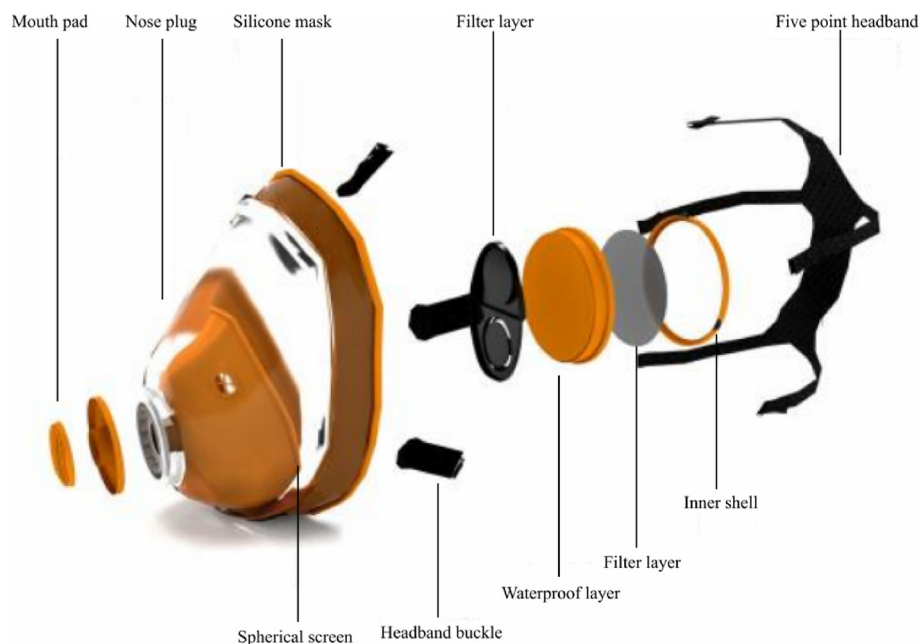


Fig. 21. Schematic diagram of breathing respiratory mask structure.

in confined spaces.

The RULA, OWAS, and Comfort Analysis for four critical task postures served as the foundation of the design. Furthermore, the design was further optimized for lightweight and comfort following ergonomics principles [30].

4.1. Analysis of operating postures

Following program improvement, the comprehensive scores of the four critical task postures of the operator in the RULA were reduced to level 2, indicating that the safety protective wearable can reasonably distribute gravity to produce a protective effect on the operator and effectively reduces the upper limb pressure load during the operation.

In the analysis of work posture, the evaluation results of the semi-squat overhaul posture and squat overhaul posture have been downgraded from level 2 and level 3 to level 1. This indicates that these postures have no adverse effects on the musculoskeletal system of the workers, thus verifying the effectiveness of the design scheme.

Additionally, the comfort level of the worker's postures were evaluated using JACK software. The bar charts in the comfort assessment are all presented in green, indicating that all the worker's body parts are in comfortable state.

4.2. Human factor analysis

To further enhance the portability and comfort of safety protective wearable, polyester and lightweight alloy are chosen as the major materials, as they are lightweight, cost-effective, and consistent with the design philosophy of lightweight, economy, and safety. Simultaneously, structural lightweight design was implemented based on the research findings of Rieffer et al. [31], with attention to spinal weight-bearing impacts on workers. A backpack structure was designed with the waist as the support point and dual shoulder support points, effectively dispersing weight, achieving lightweight, and reducing spinal load. In the design of the breathing respiratory, consideration was given to its comfort and reliability, and designs such as large-area spherical screens, large valve breathing valves, and five-point headbands were adopted.

The overall design of the wearable includes five hanging locations to address the issue of operator comfort. Furthermore, the removable shoulder padding on the back of the shoulders, hanging loops on the waist, and adjustable breathable cushioning on the legs all contribute to improved comfort and safety.

Although this study achieved some results, there is still an opportunity for future research. Future studies can focus on the following aspects.

- (1) Extend application scenarios: Apply the design method and concept of this study to the design of safety protective wearables in other similar confined work environments to verify its universality and effectiveness.
- (2) Improve usage scenarios and conduct personalized research: Due to time constraints, we mainly evaluated the assessment of four critical task postures in general scenarios. Future research can consider all possible changes in the work environment and individual differences in job habits among workers.
- (3) Expand the application of Ergonomics: investigate the application of human factors ergonomics in the design of safety protective wearables for work in confined spaces, as well as other design ideas to increase operational safety and comfort.
- (4) Enhance technology integration and innovation: Enhance technology integration and innovation by combining new technologies, such as artificial intelligence and the Internet of Things, to upgrade and innovate the design of safety protective wearable to meet the higher-level security protection needs.

5. Conclusions

This study focuses on the design of safety protective wearable for confined space operations. It deeply analyzed the potential hazards of confined space operations, pays attention to occupational health issues, and conducts research and analysis on the current development status of safety protective wearables. The potential demands and behavioral elements of users are acquired through field research and user interviews, and the STAMP - Human factors control model for safety protective wearable in confined spaces is constructed based on these findings, with the design optimized using the JACK. Finally, a set of well-functioning safety protective wearable is designed following ergonomics principles.

- (1) We conduct in-depth research on the restricted space operating environment, analyze the potential hazards of confined space operations, consider occupational health issues, examine the current state of safety protective wearables and clarify the direction and objectives of this research.
- (2) The study and analysis confirm that the design is focused on safety protective wearables in a confined spaces, and it is effectively applied to the product design based on the current state of ergonomics principles. The field research and user interviews disclose users' expectations and demands for wearable equipment, which serves as a solid foundation for future design.
- (3) Successfully developed the STAMP-Human Factors Control model for safety protective wearable in confined spaces, which includes accident prevention and comfort at the system level and provides theoretical support for design optimization.
- (4) Perform RULA, OWAS, and Comfort Assessment were used to verify the feasibility of safety protective wearables.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Ethical statement

Our study did not require further ethics committee approval as it did not involve animal or human clinical trials and was not unethical. By the ethical principles outlined in the Declaration of Helsinki, all participants provided informed consent before participating in the study. The anonymity and confidentiality of the participants were guaranteed, and participation was completely voluntary.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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